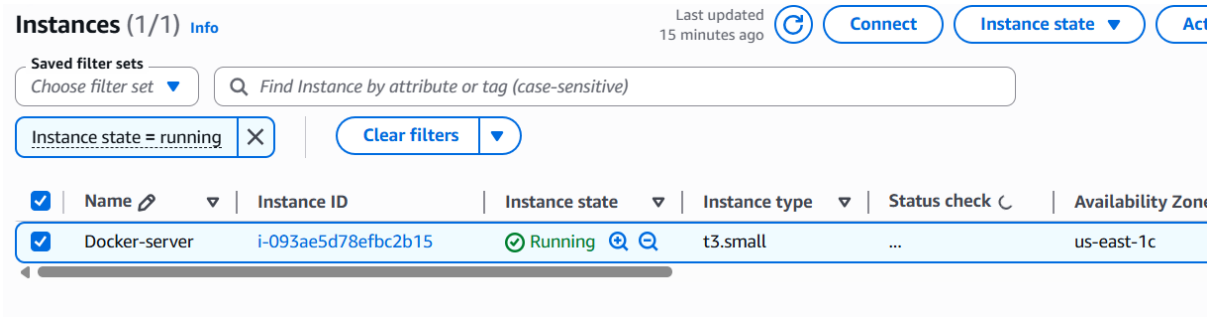


Docker-02

1. Create a tomcat container on 8080 and deploy sample application in tomcat.

- New EC2 Instance created



- Run with docker Run tomcat-app with port no 8080

```
tomcat-app
root@ip-172-31-25-147:~# docker run -d --name tomcat-app -p 8080:8080 tomcat:9
Unable to find image 'tomcat:9' locally
9: Pulling from library/tomcat
4f4fb700ef54: Pull complete
ec28c6d123a5: Pull complete
7f84b74e2ac9: Pull complete
5655393bec7b: Download complete
d97146f9bf8a: Download complete
Digest: sha256:5f08b8b793ed584f401006fa5b57d1a119baf74a8307624d48005ffe279fee08
Status: Downloaded newer image for tomcat:9
bda3d03adbdb00433e97ee82edb59aafd426987a431bb906d519e0810b85b549
root@ip-172-31-25-147:~# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS
PORTS
bda3d03adbdb   tomcat:9   "catalina.sh run"        6 seconds ago   Up 5 seconds
0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp   tomcat-app
10d183411307   tomcat:latest   "catalina.sh run"        9 minutes ago   Up 9 minutes
0.0.0.0:8081->8080/tcp, [::]:8081->8080/tcp   tomcat_latest
```

- Docker exec -it tomcat-app bash get to use for webapps to install sample war under webapps

```
root@ip-172-31-25-147:~# docker exec -it tomcat-app bash
root@bda3d03adbdb:/usr/local/tomcat# cd /usr/local/tomcat/webapps
root@bda3d03adbdb:/usr/local/tomcat/webapps# apt update && apt install wget -y
Get:1 http://archive.ubuntu.com/ubuntu noble InRelease [256 kB]
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [48.4 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [2,081 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [3,683 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1,511 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble/restricted amd64 Packages [117 kB]
```

- Installed Sample war file

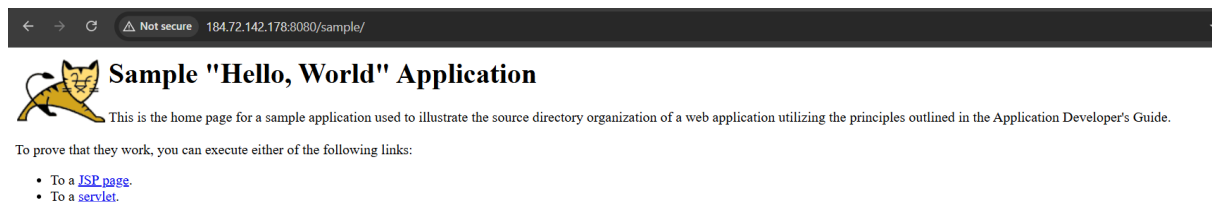
```
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
root@bda3d03adbbd:/usr/local/tomcat/webapps# wget https://tomcat.apache.org/tomcat-9.0-doc/appdev/sample/sample.war
--2026-06-01 10:07:19-- https://tomcat.apache.org/tomcat-9.0-doc/appdev/sample/sample.war
Resolving tomcat.apache.org (tomcat.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to tomcat.apache.org (tomcat.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 4606 (4.5K)
Saving to: 'sample.war'

sample.war      100%[=====>]  4.50K  --.-KB/s   in 0s
2026-06-01 10:07:19 (24.5 MB/s) - 'sample.war' saved [4606/4606]
```

```
2026-06-01 10:07:19 (24.5 MB/s) - 'sample.war' saved [4606/4606]

root@bda3d03adbbd:/usr/local/tomcat/webapps# ls
sample sample.war
root@bda3d03adbbd:/usr/local/tomcat/webapps#
```

- Check in URL with public id and 8080 port no



- Checked logs

```
root@bda3d03adbbd:/usr/local/tomcat# cd logs/
root@bda3d03adbbd:/usr/local/tomcat/logs# ls
catalina.2026-06-01.log      localhost_access_log.2026-06-01.txt
host-manager.2026-06-01.log  manager.2026-06-01.log
localhost.2026-06-01.log
root@bda3d03adbbd:/usr/local/tomcat/logs#
```

2. Create volume and deploy tomcat container on port 8081.

- Open docker ps -a and check containers 8081 is there are not and create volume and run 8081 port

```
ubuntu@ip-172-31-25-147:~$ docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS
bda3d03adbbd   tomcat:9 "catalina.sh run"        21 minutes ago    Up 21 minutes    0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp
3809334d3744   jenkins/jenkins "usr/bin/tini -- /u..." 5 days ago    Exited (143) 4 days ago
32cb6d249f26   nginx     "/docker-entrypoint...." 5 days ago    Exited (0) 4 days ago
2e28d85c0dd8   hello-world "/hello"                 5 days ago    Exited (0) 5 days ago
ubuntu@ip-172-31-25-147:~$ docker container ps
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS
bda3d03adbbd   tomcat:9 "catalina.sh run"        21 minutes ago    Up 21 minutes    0
ubuntu@ip-172-31-25-147:~$ docker run -d \
--name tomcat-deployment \
-p 8081:8080 \
-v deployment:/usr/local/tomcat/webapps \
tomcat:9
69c9be4a539738edd43e0ea9a65955a453f49c042ccc8cfb75f50c34d9fae15c
```

- Check docker ps and 8081 created and created volume open and go to webapps

```
ubuntu@ip-172-31-25-147:~$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        P
ORTS
69c9be4a5397   tomcat:9 "catalina.sh run"       10 seconds ago Up 9 seconds   0
.0.0.0:8081->8080/tcp, [::]:8081->8080/tcp tomcat-deployment
bda3d03adbdb   tomcat:9 "catalina.sh run"       21 minutes ago Up 21 minutes   0
.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp tomcat-app
```

```
ubuntu@ip-172-31-25-147:~$ docker exec -it tomcat-deployment bash
root@69c9be4a5397:/usr/local/tomcat# cd /usr/local/tomcat/webapps
root@69c9be4a5397:/usr/local/tomcat/webapps# ls
root@69c9be4a5397:/usr/local/tomcat/webapps# pwd
/usr/local/tomcat/webapps
```

- Now install Sample war file here with URL after installed inspect

```
Processing triggers for libc-bin (2.35-0ubuntu3.7) ...
root@69c9be4a5397:/usr/local/tomcat/webapps# wget https://tomcat.apache.org/tomcat-9.0-doc/appdev/sample/sample.war
--2026-06-01 10:29:39-- https://tomcat.apache.org/tomcat-9.0-doc/appdev/sample/sample.war
Resolving tomcat.apache.org (tomcat.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to tomcat.apache.org (tomcat.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 4606 (4.5K)
Saving to: 'sample.war'

sample.war          100%[=====>]  4.50K  --.-KB/s   in 0s
```

```
2026-06-01 10:29:39 (37.8 MB/s) - 'sample.war' saved [4606/4606]

root@69c9be4a5397:/usr/local/tomcat/webapps# ls
sample sample.war
root@69c9be4a5397:/usr/local/tomcat/webapps#
```

- Inspect volume here the location is visible clearly

```
root@69c9be4a5397:/usr/local/tomcat/webapps# exit
exit
ubuntu@ip-172-31-25-147:~$ docker inspect tomcat-deployment
[
  {
    "Id": "69c9be4a539738edd43e0ea9a65955a453f49c042ccc8cfb75f50c34d9fae15c",
    "Created": "2026-06-01T10:28:14.758869216Z",
    "Path": "catalina.sh",
    "Args": [
      "run"
    ],
    "State": {
      "Status": "running",
      "Running": true,
      "Paused": false,
```

```
    },
    "Mounts": [
      {
        "Type": "volume",
        "Name": "deployment",
        "Source": "/var/lib/docker/volumes/deployment/_data",
        "Destination": "/usr/local/tomcat/webapps",
        "Driver": "local",
        "Mode": "z",
        "RW": true,
        "Propagation": ""
      }
    ]
  }
]
```

- Open browser with public id and port no 8081



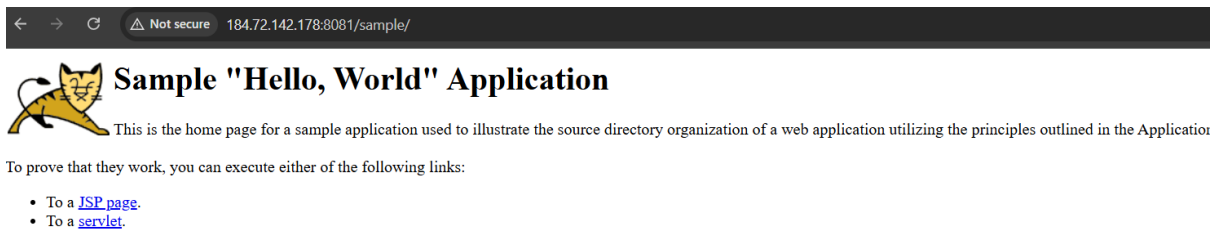
To prove that they work, you can execute either of the following links:

- To a [JSP page](#).
- To a [servlet](#).

- Now remove the volume and run again it will run again

```
ubuntu@ip-172-31-25-147:~$ docker rm -f tomcat-deployment
tomcat-deployment
ubuntu@ip-172-31-25-147:~$ docker run -d \
--name tomcat-deployment \
-p 8081:8080 \
-v deployment:/usr/local/tomcat/webapps \
tomcat:9
3ebf822dcf1992c2a137bb181c7698a645e95360272dbfe54099e5c7b3f7755e
ubuntu@ip-172-31-25-147:~$
```

- Check it again the URL working



To prove that they work, you can execute either of the following links:

- To a [JSP page](#).
- To a [servlet](#).

3. Limit the nginx container to 500 MB.

- Docker stats first checked the size of containers

```
ubuntu@ip-172-31-25-147: ~
CONTAINER ID   NAME             CPU %     MEM USAGE / LIMIT   MEM %     NET I/O       BLOCK I/O     PIDS
3ebf822dcf19   tomcat-deployment 0.18%     79.32MiB / 1.861GiB  4.16%     1.95kB / 755B  270kB / 106kB  35
bda3d03adbdb   tomcat-app        0.11%     85.82MiB / 1.861GiB  4.50%     39MB / 70.6kB  1.39MB / 105MB  35
```

- Now pull nginx and given 500mb size limit

```
ubuntu@ip-172-31-25-147:~$ docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
Digest: sha256:5aca99593157f4ae539a5dec1092a0ad8762f8e2eb1789085a13a0f5622369f6
Status: Image is up to date for nginx:latest
docker.io/library/nginx:latest
ubuntu@ip-172-31-25-147:~$ docker container run -itd -p 8082:8080 -m 500m nginx
20808c85d69f892b492c3b23339aa4831e94d9597db9813f01578493a0e73a12
```

- After that again check docker stats

```
ubuntu@ip-172-31-25-147:~$ docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
Digest: sha256:5aca99593157f4ae539a5dec1092a0ad8762f8e2eb1789085a13a0f5622369f6
Status: Image is up to date for nginx:latest
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O     PIDS
ea9ced9662df   inspiring_edison  0.00%    14.64MiB / 500MiB      2.93%    1.09kB / 126B   13.4MB / 12.3kB   3
3ebf822dcf19   tomcat-deployment  0.26%    79.76MiB / 1.86GiB     4.19%    2.59kB / 1.19kB 270kB / 340kB     35
bda3d03adbdb   tomcat-app       0.10%    86.25MiB / 1.86GiB     4.53%    39MB / 72.2kB   1.39MB / 105MB    35
```

4. Create a sample docker file using below instructions.

1. Base module as amazonlinux: latest
2. Maintainer you name
3. Install nginx
4. COPY one index.html file to image
5. Expose on port 80
6. Command to start the nginx container

- Create a new Vi docker file and index.html file

```
root@ip-172-31-25-147: ~/docker-nginx
root@ip-172-31-25-147:~/docker-nginx# vi Dockerfile
root@ip-172-31-25-147:~/docker-nginx# vi Dockerfile
root@ip-172-31-25-147:~/docker-nginx# ls
Dockerfile  docker-nginx  index.html
```

```
FROM amazonlinux:latest
LABEL maintainer="Arunasai"
RUN yum update -y && \
    yum install nginx -y
COPY index.html /usr/share/nginx/html/index.html
EXPOSE 80
CMD ["nginx", "-g", "daemon off;"]
```

- After done with file Build it

```
Dockerfile  docker-nginx  index.html
root@ip-172-31-25-147:~/docker-nginx# docker build -t custom-nginx:v1 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  3.584kB
Step 1/6 : FROM amazonlinux:latest
latest: Pulling from library/amazonlinux
c7345fb8c651: Pulling fs layer
b3d837a98232: Download complete
2da5356a79d8: Download complete
c7345fb8c651: Download complete
c7345fb8c651: Pull complete
Digest: sha256:10922927c68854781965c63232388c6071c302313140617104677c632
```

- Successfully build and check images

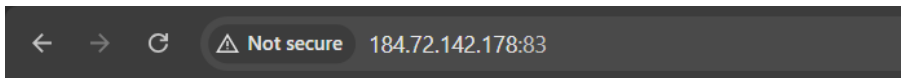
```
Complete!
--> Removed intermediate container 556e955232fe
--> aa39f8293916
Step 4/6 : COPY index.html /usr/share/nginx/html/index.html
--> c60fe29622fc
Step 5/6 : EXPOSE 80
--> Running in 1a7c942e77f8
--> Removed intermediate container 1a7c942e77f8
--> 2cb919b944ed
Step 6/6 : CMD ["nginx", "-g", "daemon off;"]
--> Running in fa399fbd655f
--> Removed intermediate container fa399fbd655f
--> 925917d788e0
Successfully built 925917d788e0
Successfully tagged custom-nginx:v1
root@ip-172-31-25-147:~/docker-nginx#

root@ip-172-31-25-147:~/docker-nginx# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
amazonlinux:latest	ad882807c6c8	227MB	59.3MB	
custom-nginx:v1	925917d788e0	584MB	195MB	
hello-world:latest	0e760fdfbc48	25.9kB	9.49kB	U
jenkins/jenkins:latest	d566c401e890	801MB	290MB	U
nginx:latest	5aca99593157	241MB	66MB	U
sonarqube:latest	35bedac3f40c	2.52GB	1.08GB	
tomcat:9	5f08b8b793ed	592MB	158MB	U
tomcat:latest	247650787b97	589MB	158MB	
ubuntu:latest	f3d28607ddd7	160MB	45.3MB	

```
root@ip-172-31-25-147:~/docker-nginx#
```

- Check with URL Nginx working successfully

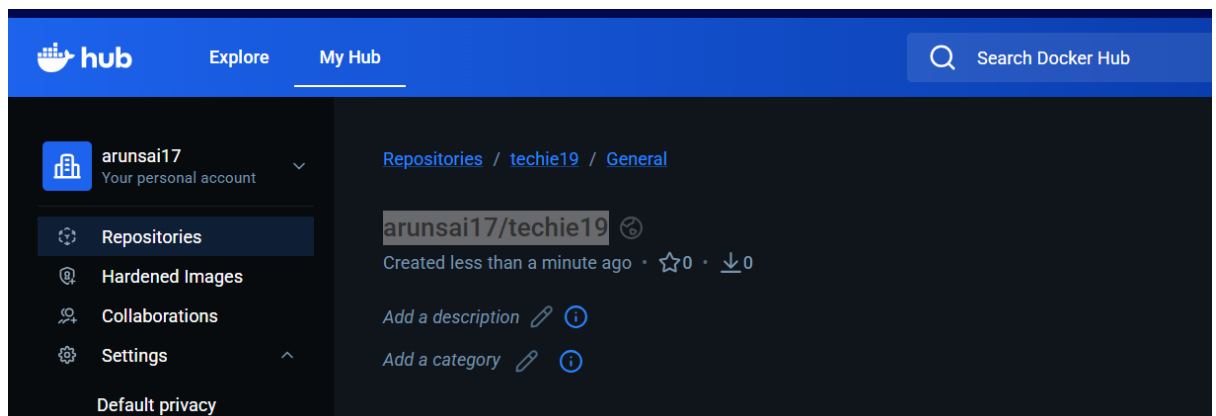


Welcome to Docker Nginx

Created by Arun Sai

5. Push image to Docker hub

- Created new Docker hub and created new repository



- Check docker images and change the name

```
root@ip-172-31-25-147:~/docker-nginx# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
amazonlinux:latest	ad882807c6c8	227MB	59.3MB	
custom-nginx:v1	925917d788e0	584MB	195MB	U
hello-world:latest	0e760fdfbc48	25.9kB	9.49kB	U
jenkins/jenkins:latest	d566c401e890	801MB	290MB	U
nginx:latest	5aca99593157	241MB	66MB	U
sonarqube:latest	35bedac3f40c	2.52GB	1.08GB	
tomcat:9	5f08b8b793ed	592MB	158MB	U
tomcat:latest	247650787b97	589MB	158MB	
ubuntu:latest	f3d28607ddd7	160MB	45.3MB	

```
root@ip-172-31-25-147:~/docker-nginx# docker tag custom-nginx:v1 arunsail7/techie19:v1
root@ip-172-31-25-147:~/docker-nginx# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
amazonlinux:latest	ad882807c6c8	227MB	59.3MB	
arunsail7/techie19:v1	925917d788e0	584MB	195MB	U
custom-nginx:v1	925917d788e0	584MB	195MB	U
hello-world:latest	0e760fdfbc48	25.9kB	9.49kB	U
jenkins/jenkins:latest	d566c401e890	801MB	290MB	U
nginx:latest	5aca99593157	241MB	66MB	U
sonarqube:latest	35bedac3f40c	2.52GB	1.08GB	
tomcat:9	5f08b8b793ed	592MB	158MB	U
tomcat:latest	247650787b97	589MB	158MB	
ubuntu:latest	f3d28607ddd7	160MB	45.3MB	

- Login with docker hub with URL and password

```
root@ip-172-31-25-147:~/docker-nginx# docker login
```

USING WEB-BASED LOGIN

Info → To sign in with credentials on the command line, use 'docker login -u <username>'

Your one-time device confirmation code is: RCMS-RMQN
 Press ENTER to open your browser or submit your device code here: <https://login.docker.com/activate>

Waiting for authentication in the browser...

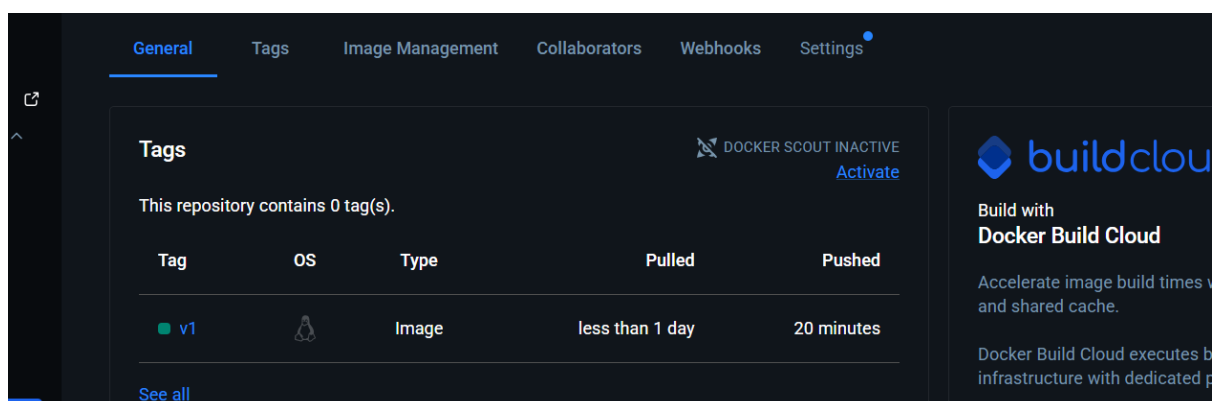
WARNING! Your credentials are stored unencrypted in '/root/.docker/config.json'.
 Configure a credential helper to remove this warning. See
<https://docs.docker.com/go/credential-store/>

Login Succeeded

- Pushed to Docker hub successfully

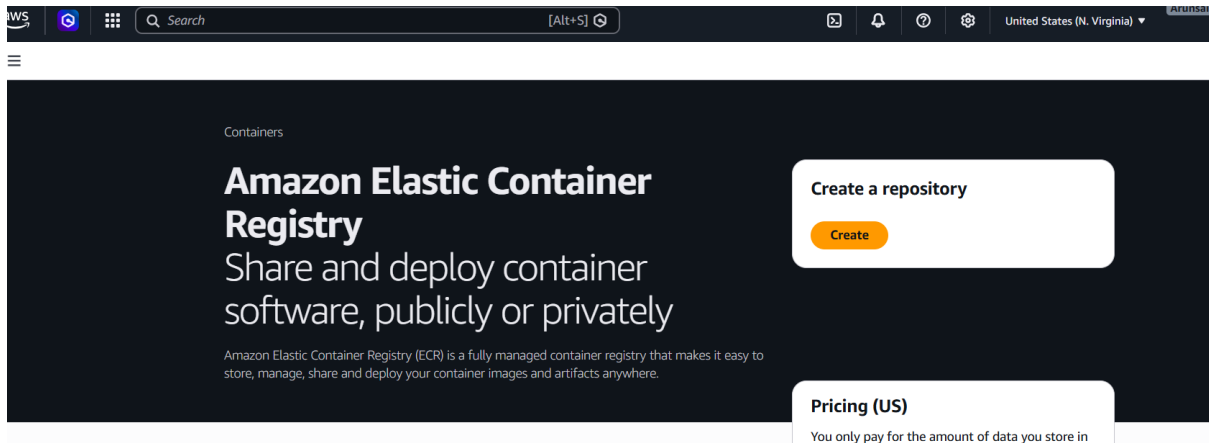
```
root@ip-172-31-25-147:~/docker-nginx# docker push arunsail7/techie19:v1
The push refers to repository [docker.io/arunsail7/techie19]
10d3dbd157fe: Pushed
c7345fb8c651: Mounted from library/amazonlinux
bcf200246380: Pushed
v1: digest: sha256:925917d788e011a8b695385588af86d454c8da06ad703ef598d08392671ebb6d size: 972
```

- Checked in Docker Hub successfully

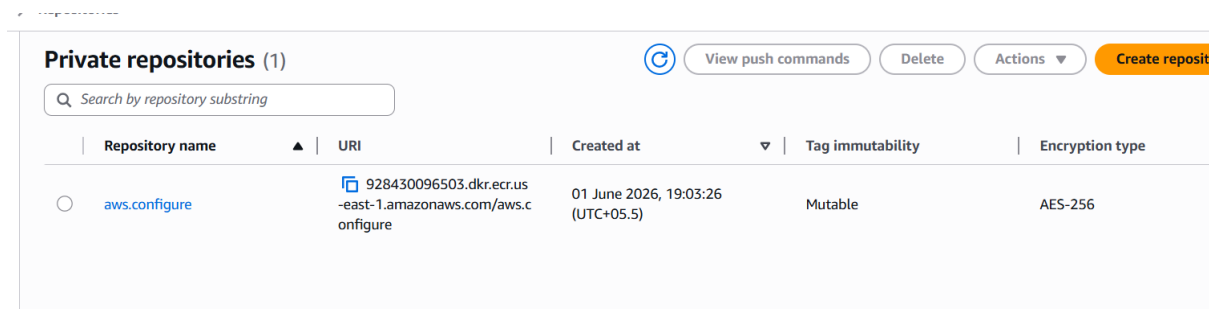


6. push image to AWS ECR

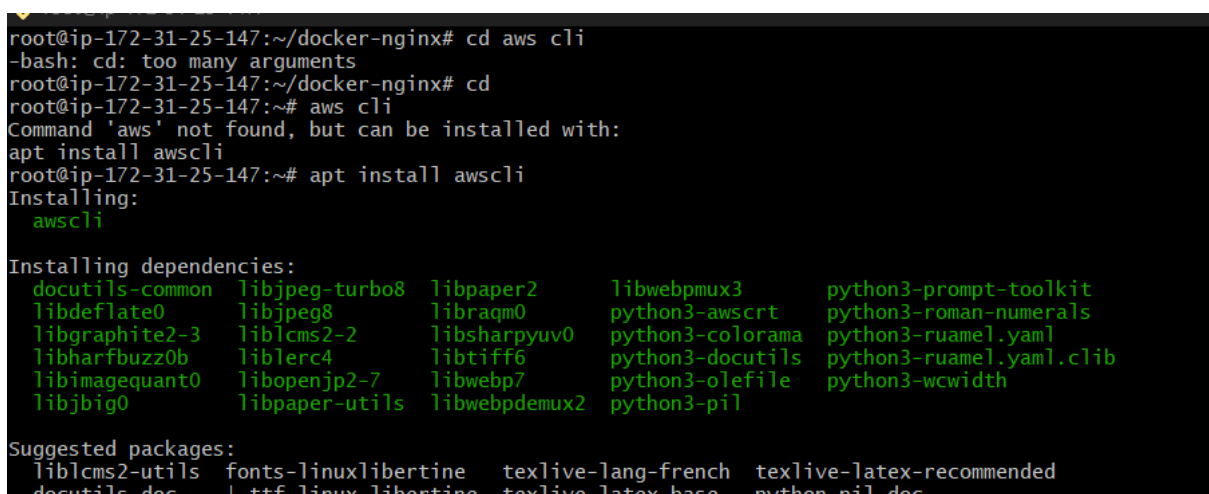
- Open AWS ECR and create Repo



- Created new repo



- Login with AWS configure cli



- After login with secret details and login to docker

```
root@ip-172-31-25-147:~# aws configure
AWS Access Key ID [None]: AKIA5QKV7AB3RFRZF7W2
AWS Secret Access Key [None]: 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
Default region name [None]: us-east-1
Default output format [None]: json
root@ip-172-31-25-147:~# aws ecr get-login-password --region us-east-1 | \
docker login \
--username AWS \
--password-stdin 928430096503.dkr.ecr.us-east-1.amazonaws.com/aws.configure

WARNING! Your credentials are stored unencrypted in '/root/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
root@ip-172-31-25-147:~# docker images
```

- Check docker images

```
root@ip-172-31-25-147:~# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
amazonlinux:latest	ad882807c6c8	227MB	59.3MB	
arunsail7/techie19:v1	925917d788e0	584MB	195MB	
custom-nginx:v1	371e84608aa7	584MB	195MB	U
hello-world:latest	0e760fdfbc48	25.9kB	9.49kB	U
jenkins/jenkins:latest	d566c401e890	801MB	290MB	U
nginx:latest	5aca99593157	241MB	66MB	U
sonarqube:latest	35bedac3f40c	2.52GB	1.08GB	
tomcat:9	5f08b8b793ed	592MB	158MB	U
tomcat:latest	247650787b97	589MB	158MB	
ubuntu:latest	f3d28607ddd7	160MB	45.3MB	

- Now push docker to repo with tag

```
root@ip-172-31-25-147:~# docker tag custom-nginx:v1 928430096503.dkr.ecr.us-east-1.amazonaws.com/aws.configure
root@ip-172-31-25-147:~# docker push 928430096503.dkr.ecr.us-east-1.amazonaws.com/aws.configure
Using default tag: latest
The push refers to repository [928430096503.dkr.ecr.us-east-1.amazonaws.com/aws.configure]
93a064f8c941: Pushed
c7345fb8c651: Pushed
bcf200246380: Pushed
latest: digest: sha256:371e84608aa78f1a29a9e310e6f485fff9b5700aaba2c4b8f2497dac153829f8 size: 972
root@ip-172-31-25-147:~#
```

- In Repo we can see successfully pushed

aws.configure

Summary | **Images** | Lifecycle policy | Permissions | Repository tags

Images (1) Info 🔄 Delete 📄 Copy URI Details Scan View push commands

🔍 Filter active images

☐	Image tags	Type	Created at	Image size (MB)	Image digest	Last pulled at
☐	latest	Image	01 June 2026, 19:09:32 (UTC+05.5)	195.48	sha256:371e846...	-